

Course Handout

Institute/School Name	Chitkara University Institute of Engineering and Technologys		
Department Name	Computer Science & Engineering		
Programme Name	Bachelor of Engineering (B.E) - Computer Science & Engineering		
Course Name	OPERATING SYSTEM AND LINUX FUNDAMENTALS	Session	Jan 2026-June 2026
Course Code	25CSE0104	Semester/Batch	2nd/2025
L-T-P (Per Week)	2-0-2	Course Credits	3
Prerequisite	NA	NHEQF Level	4.5
Course Coordinator	Dr. Chander Prabha	SDG Number	4, 9

1. Objectives of the Course

This course focuses on fundamental design and implementation ideas in the engineering of operating systems. Topics include virtual memory, threads, context switches, kernels, interrupts, system calls, interprocess communication, coordination, and the interaction between software and hardware. The main objectives of the course are to:

- Impart an inclusive and structured approach to understanding the concepts of operating systems, ensuring equitable and high-quality learning opportunities.
- Expose students to concurrent programming issues in the management of resources like processors, memory, and input-output, fostering skills for innovative and sustainable technological advancements.
- Provide practical skills required as a foundation to build solutions for real-world engineering problems, supporting sustainable infrastructure and resilient technology systems.
- Enable learners to understand and apply diverse Linux operating commands effectively, promoting technological literacy and innovation in line with sustainable development goals.

2. Course Learning Outcomes (CLOs)

Student should be able to:

	CLOs	Program Outcomes (PO)	NHEQF Level Description	No. of Lectures
CLO01	Understand the role and structure of operating systems and analyze their components, including process, memory, storage, and I/O management.	PO1, PO3, PO11	Q1, Q2	10
CLO02	Explain process management concepts, including process scheduling, inter-process communication, and threading models, for efficient resource utilization.	PO1, PO3, PO11	Q3, Q6	12
CLO03	Evaluate CPU scheduling algorithms to determine their suitability based on scheduling criteria and system requirements, and methods for handling deadlock	PO1, PO4, PO5, PO11	Q1, Q2	18
CLO04	Analyze synchronization mechanisms and resolve synchronization issues using semaphores, monitors, and critical section solutions.	PO3, PO4, PO5, PO11	Q3, Q6	6
CLO05	Examine memory management techniques, such as paging, segmentation, and contiguous memory allocation, and their impact on system performance. File Management and Mass Storage Structure.	PO3, PO4, PO5, O11	Q3	12
CLO06 (Only for lab components)	Design, implement, and analyze key operating system functionalities, including process management, CPU scheduling, process synchronization, deadlock handling, and memory management, through practical programming exercises.	PO1, PO3, PO4, PO5, PO11	Q3, Q6	6
Total Contact Hours				64

CLO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	Type of Assessment
CLO01	H		H								H	L		Formative, Summative
CLO02	M		H								H	L	L	Formative, Summative
CLO03	M			H	H			H			M	M	M	Formative, Summative
CLO04			H	M	L				L		L	M	M	Formative, Summative
CLO05			H	M	H						L		H	Formative, Summative
CLO06	M		H	H	H						H	H	H	Formative

H=High, M=Medium, L=Low

3. Recommended Books:

B01: Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, and Greg Gagne, 9th Edition, 2018, Wiley.

B02: System Programming and Operating Systems, D.M. Dhamdhare, 2nd Edition, 2009, Tata McGraw Hill.

B03: Operating Systems: Internals and Design Principles, William Stallings, 9th Edition, 2018, Pearson.

B04: Modern Operating Systems, Andrew S. Tanenbaum, 4th Edition, 2016, Pearson.

B05: Advanced Linux Programming, Mark Mitchell, Jeffrey Oldham, and Alex Samuel, 1st Edition, 2001 Germany: New Riders.

B06: Linux Command Line and Shell Scripting Bible, Richard Blum and Christine Bresnahan, 3rd Edition, 2015, Wiley.

4. Other readings and relevant websites:

Serial No	Link of Journals, Magazines, websites and Research Papers
1	http://nptel.ac.in/courses/106108101/
2	http://www.ics.uci.edu/~ics143/lectures.html
3	http://www.cs.kent.edu/~farrell/osf03/oldnotes/index.html
4	http://williamstallings.com/OS/OS6e.html
5	https://learning.edx.org/course/course-v1:LinuxFoundationX+LFS101x+1T2017
6	https://www.gnu.org/software/bash/manual/
7	https://ocw.mit.edu/courses/6-828-operating-system-engineering/
8	https://nptel.ac.in/courses/106106144

5. Recommended Tools and Platforms

VirtualBox 6.1.22, Ubuntu 21.04, GCC Compiler

6. Course Plan: Theory+ Lab Theory

Plan

Lect. No.	Topic(s)
1-2	Introduction: Introduction to Operating systems, Operating System role, Computer system architecture: Single processor, Multiprocessor Systems, Clustered Systems, OS operations, Components of OS: Process management, memory management, storage management, I/O management
3-4	OS Structure: Monolithic, Layered, Microkernel, Kernel, Shell, OS services and system calls, User and OS interface, System Calls/API, Types of System Calls, System Programs
5-6	Introduction to Linux OS: History, philosophy, distributions, Linux architecture overview.
7-8	Process Management

	Process Concept: Process definition, PCB, process states, state transition diagram. Process Operations: Process creation, termination, context switching, IPC basics.
9-10	Threads: Overview of Threads, Multicore Programming, Multithreading Models, Threading Issues, Linux Threads
11-13	CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithm-, First in First Out Scheduling Algorithms (FIFO), Shortest Job First Scheduling Algorithms (SJF)
14-15	Priority Scheduling Algorithms, Scheduling Algorithms- Round-robin Scheduling Algorithms, Multilevel Queue Scheduling
ST1 (0 to 45%)	
16-18	Process Synchronization: Background, The Critical-Section Problem, Two-process solution, Multiple-Process solution, Synchronization Hardware. Semaphores
19-21	Deadlock: System Model Deadlock Characterization, Methods for handling Deadlocks, Deadlock Prevention, Deadlock avoidance, Deadlock detection, Recovery from Deadlocks
22-25	Memory Management and Virtualisation Memory Management: Logical vs physical address space, MMU, swapping, Contiguous memory allocation, Fragmentation, Segmentation, Paging, Segmentation with paging
26-29	Virtual Memory: Introduction, Demand Paging, Page Replacement Algorithms, Allocation of frames, Thrashing
ST2 (46% to 90%)	
30-31	Mass Storage Structure: Overview, Disk Structure, Disk Attachment, Disk Scheduling, Disk Management
32	File Management System: File Allocation methods, free-space management.
End Term Exam (0-100%)	

Lab Plan

Lab No.	Topic(s)
1-4	a) Installation: Configuration & Customisations of Linux b) Introduction to GCC compiler: Basics of GCC, Compilation of program, Execution of program c) Time stamping, d) Automating the execution using makefile.
5-6	Basic Commands of Linux OS: Navigation (ls, cd), File manipulation (cp, mv, rm), and Viewing (cat, less), where is, what is and their options
7-8	Execute Directory-oriented commands: cd, pwd, mkdir, rmdir and their options
9-10	Execute the basic and user status commands like: su, sudo, man, help, history, who, whoami, id, uname, uptime, free, tty, cal, date, hostname, reboot, clear and their options
11-14	Write programs to create and manage processes using system calls such as fork(), exec(), and wait(). Study parent-child relationships and process lifecycle behaviour in Linux.
15-16	Write a program to implement Process concepts by printing the process ID.
17-18	Implement FCFS, SJF scheduling algorithms.
19-20	Implement Linux memory management commands: Overview, free, vmstat, Memory Allocation System Calls (malloc, sbrk). Analyse how memory is allocated and utilized by running processes.
21-22	Implement deadlock detection in C. (Banker's Algorithm)
23-26	File System Concepts: File attributes, access methods, directory structures, protection.
27-28	Execute File system commands: Comparing Files using diff, cmp, comm and their options
29-30	Execute File system <i>system-calls</i> : open, read, write, close, lseek, stat and their options
31-32	Execute Linux file management commands: chmod, chown, ln, mount, df, du and their options
Continuous Evaluation 1	

7. Delivery/Instructional Resources

Theory Plan:

Lect. No.	Topics	CLO	Book No, CH No, Page No	TLM	ALM	Web References	Audio-Video
1-2	Introduction: Introduction to Operating systems, Operating System role, Computer system architecture: Single processor, Multiprocessor Systems, Clustered Systems, OS operations, Components of OS: Process management, memory management, storage management, I/O management	CLO01	B02, CH 11.1, 11.2, Page no 412-422 B01, CH 1.1-1.8, Page no 4-29	Lecture	Quiz, Test questions	https://nptel.ac.in/courses/106108101	Introduction to Operating Systems Coursera https://nptel.ac.in/courses/106105105/106105214/ https://www.youtube.com/watch?v=BURt97EkA
3-4	OS Structure: Monolithic, Layered, Microkernel, Kernel, Shell, OS services and system calls, User and OS interface, System Calls/API, Types of System Calls, System Programs	CLO01	B01, CH 2.2-2.5, Page no 58-74 B05, CH 1, Page no 3-10	Lecture	Brainstorming Sessions, Focused Listing	https://nptel.ac.in/courses/106108101	https://www.youtube.com/watch?v=IQWERTMoKbl Introduction to Operating Systems Coursera https://nptel.ac.in/courses/106105105/106105214/
5-6	Introduction to Linux OS: History, philosophy, distributions, Linux architecture overview.	CLO01	B01, CH 2.2-2.5, Page no 58-74 B05, CH 1, Page no 3-10	Lecture	Brainstorming Sessions, Focused Listing	https://codemaniacs.tudio.com/unit-1-introduction-to-linux-operating-system-2/	https://www.youtube.com/watch?v=h_xSovNStSE
7-8	Process Management Process Concept: Process definition, PCB, process states, state transition diagram. Process Operations: Process creation, termination, context switching, IPC basics.	CLO02	B02, CH 11.1, 11.2, Page no 423-428 B01, CH 3.1-3.5, Page no 105-128 B05, CH 5.1-5.3, Page no 95-105	Lecture	Focused Listing	https://nptel.ac.in/courses/106108101	https://www.youtube.com/watch?v=OrM7nZcxXZU Introduction to Operating Systems Coursera https://nptel.ac.in/courses/106105105/106105214/



9-10	Threads: Overview of Threads, Multicore Programming, Multithreading Models, Threading Issues, Linux Threads	CLO01, CLO02	B03, CH 4.1, Page no 160-174 B01, CH 4.3-4.6, Page no 169-185 B05, CH 41.-4.4, Page no 61- 88.	Lecture	Brainstorming Sessions	https://nptel.ac.in/courses/106108101	https://www.youtube.com/watch?v=L0FGJcVnvAk Introduction to Operating Systems Coursera https://nptel.ac.in/courses/106105/106105214/
11-13	CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithm-, First in First Out Scheduling Algorithms (FIFO), Shortest Job First Scheduling Algorithms (SJF)	CLO02, CLO03	B01, CH 6.1-6.6, Page no 261-290	Lecture	Leading Question, Quiz, Brainstorming Sessions	https://nptel.ac.in/courses/106108101	https://www.youtube.com/watch?v=bWHFY8-rLSI&list=PLBlnK6fEYqRiVhbXDG_LXDk_OQAEuVcp2O&index=40 Introduction to Operating Systems Coursera https://nptel.ac.in/courses/106105/106105214/
14-15	Priority Scheduling Algorithms, Scheduling Algorithms- Round-robin Scheduling Algorithms, Multilevel Queue Scheduling	CLO02, CLO03	B03, CH 9.1-9.2, Page no 405-430	Lecture	Quiz, Peer Review	https://nptel.ac.in/courses/106108101	Introduction to Operating Systems Coursera https://nptel.ac.in/courses/106105/106105214/
16-17	Process Synchronization: Background, The Critical-Section Problem, Two- process solution, Multiple-Process solution,	CLO04	B01, CH 5.1-5.4, Page no 203-210	Lecture	Quiz, Peer Review, Brainstorming Sessions	https://nptel.ac.in/courses/106108101	Introduction to Operating Systems Coursera https://nptel.ac.in/courses/106105/106105214/ https://www.youtube.com/watch?v=ph2awKa8r5Y&list=PLBlnK6fEYqRiVhbXDG_LXDk_OQAEuVcp2O&index=57
18	Synchronization Hardware, Semaphores.	CLO03, CLO04	B03, CH 5.1-5.4, Page no 207-232	Lecture,	Quiz, Peer Review	https://nptel.ac.in/courses/106108101	Introduction to Operating Systems Coursera https://nptel.ac.in/courses/106105/106105214/



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19-21	Deadlock: System Model Deadlock Characterization, Methods for handling Deadlocks, Deadlock Prevention, Deadlock avoidance, Deadlock detection, Recovery from Deadlocks	CLO04	B01, CH 7.1-7.7, Page no 315-338	Lecture	Quiz, Peer Review	https://nptel.ac.in/courses/106108101	Introduction to Operating Systems Coursera https://nptel.ac.in/courses/106/105/106105214/ https://www.youtube.com/watch?v=7bnpFpYZtVvk&list=PLBlnK6fEvqRiVhbxXDGLXk_OQAeuVcp2O&index=73
22-25	Memory Management and Virtualisation Memory Management: Logical vs physical address space, MMU, swapping, Contiguous memory allocation, Fragmentation, Segmentation, Paging, Segmentation with paging	CLO05	B02, CH 12.1-12.5 Page no 434-460	Lecture	Quiz, Peer Review	https://nptel.ac.in/courses/106108101	Introduction to Operating Systems Coursera https://nptel.ac.in/courses/106/105/106105214/ https://www.youtube.com/watch?v=LKYKp_ZzlvM https://www.youtube.com/watch?v=d9WverblWQc&list=PLBlnK6fEvqRiVhbxXDGLXk_OQAeuVcp2O&index=74
26-29	Virtual Memory: Introduction, Demand Paging, Page Replacement Algorithms, Allocation of frames, Thrashing	CLO05	B02, CH 13.1-12.5 Page no 461-489	Lecture	Quiz, Peer Review	https://www.bbau.ac.in/dept/dit/TM/Virtual%20Memory%20and%20Demand%20Paging.pdf	https://www.youtube.com/watch?v=QSVInZRS4og
30-31	Mass Storage Structure:	CLO05	B03, CH 14.1-12.7,	Lecture	Quiz, Peer Review	https://nptel.ac.in/courses/106108101	Introduction to Operating Systems



	Overview, Disk Structure, Disk Attachment, Disk Scheduling, Disk Management		Page no 521-579				Coursera https://nptel.ac.in/courses/106/105/106105214/ https://www.youtube.com/watch?v=syq56cLrWdI&list=PLBlNk6fEvgRiVhbXDGLXDKOQAeuVcp2O&index=78
32	File Management System File Allocation methods, free-space management.	CLO05	B03, CH 15.1-12.7, Page no 589-621	Lecture	Quiz, Peer Review	https://aissmspoly.org.in/wp-content/uploads/2020/01/File-management-.pdf	https://www.youtube.com/watch?v=DYGIPiez5GY&list=PLYSKRj7KSLYgS9jVvonZTFpiYkizZXF2-

Lab Plan:

Lab No.	Experiment	CLO	TLM	ALM	Web References	Audio-Video
1-4	a) Installation: Configuration & Customisations of Linux b) Introduction to GCC compiler: Basics of GCC, Compilation of program, Execution of program c) Time stamping, d) Automating the execution using makefile	CLO01	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	https://ubuntu.com/tutorials/install-ubuntu-server#1-overview	https://www.youtube.com/watch?v=oLjN6jAg-sY
5-6	Basic Commands of Linux OS: Navigation (ls, cd), File manipulation (cp, mv, rm), and Viewing (cat, less), where is, what is and their options	CL006	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	http://tldp.org/LDP/abs/html/basics.html	https://www.youtube.com/watch?v=YHFzrakOas
7-8	Execute Directory-oriented commands: cd, pwd, mkdir, rmdir and their options	CLO06	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	http://litux.nl/Reference/Books/7213/du0082.html	https://www.youtube.com/watch?v=YHFzrakOas



9-10	Execute the basic and user status commands like: su, sudo, man, help, history, who, whoami, id, uname, uptime, free, tty, cal, date, hostname, reboot, clear and their options	CLO04	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	https://www.geeksforgeeks.org/linux-unix/basic-linux-commands/	https://www.youtube.com/watch?v=2iVv1LKSGSM&t=8s
11-14	Write programs to create and manage processes using system calls such as fork(), exec(), and wait(). Study parent-child relationships and process lifecycle behaviour in Linux.	CLO04	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	https://ies.in/blog/understanding-fork-exec-and-process-creation-in-linux/	https://www.youtube.com/watch?v=XG1ptxr6O34
15-16	Implement Process concepts using C language by Printing process Id.	CLO02, CLO03	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	https://linuxhint.com/fork-system-call-linux/	https://www.youtube.com/watch?v=oLjN6jAg-sY
17-18	Implement FCFS, SJF scheduling algorithms.	CLO03	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	https://www.tutorialspoint.com/c-program-for-fcfs-scheduling https://www.javatpoint.com/round-robin-program-in-c	https://www.youtube.com/watch?v=mSwS1QLpdB8
19-20	Execute Linux memory management commands: Overview, free, vmstat, Memory Allocation System Calls (malloc, sbrk). Analyze how memory is allocated and utilized by running processes.	CLO03	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	https://minnie.tuhs.org/CompArch/Labs/week9.html https://www.cse.iitb.ac.in/~mythili/os/iitb_slides/memory-syscall.pdf	https://www2.cs.uregina.ca/~hamilton/courses/330/notes/unix/memory/memory.html
21-22	Implement deadlock detection (Banker's Algorithm)	CLO04	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	https://www.codingninjas.com/codestudio/library/deadlock-detection-algorithm	https://www.youtube.com/watch?v=XVZMxXB

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23-26	File System Concepts: File attributes, access methods, directory structures, protection.	CLO04	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	https://aismspoly.org.in/wp-content/uploads/2020/01/File-management-.pdf	https://www.youtube.com/watch?v=DYGIPiez5GY&list=PLySKRj7KSLYgS9jVyonZTFpiYklzZXf2-
27-28	Execute File system commands: Comparing Files using diff, cmp, comm and their options	CLO06	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	https://www.geeksforgeeks.org/cmp-command-in-linux-with-examples/ https://www.geeksforgeeks.org/diff-command-linux-examples/	https://www.youtube.com/watch?v=YHFzrakOas
29-30	Execute File system <i>system-calls</i> : open, read, write, close, lseek, stat and their options	CLO06	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	https://www.geeksforgeeks.org/c/input-output-system-calls-c-create-open-close-read-write/	https://www.youtube.com/watch?v=gMAVnpLhLPo
31-32	Execute Linux file management commands: chmod, chown, ln, mount, df, du and their options	CLO06	Discussion, Demonstration method, Questioning	Brainstorming Sessions, Think / Pair / Share	https://www.stroingdm.com/blog/linux-file-permissions	https://www.youtube.com/watch?v=nE9l8oMXU1M

8. Remedial Classes

After every sessional tests, different types of learners will be identified, and special discussions will be planned and scheduled accordingly.

Action Plan for different types of learners:

Learner Type I	Learner Type II	Learner Type III
Remedial Classes, and Guided Tutorials	Doubt Sessions, Guided Tutorials	Guided Tutorials

9. Self-Learning

Assignments to promote self-learning, survey of contents from multiple sources.

S.No	Topics	CLO	ALM	References/MOOCs
1	Mobile Operating Systems Features of Android and iOS	CLO06	Think – Pair-Share	https://www.geeksforgeeks.org/operating-systems/difference-between-ios-and-android/

10. Delivery Details of Content Beyond Syllabus

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.No	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCs
1.	Cloud OS Internals: Resource Virtualization in AWS, Azure	CLO06	PO11	Shadowing	https://www.youtube.com/watch?v=cmdTnA6hN_A

11. Evaluation Scheme & Components:

Assessment Type	Evaluation Component	Type of Component	No. of Assessments	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Formative	Component 1	Continuous Lab Evaluation	01*	20%	20	Practical, Viva, file	CLO01-CLO06
Summative	Component 2	Sessional Tests (STs)	02**	30%	30	Computer-Based Test	CLO01-CLO06
Summative	Component 3	End Term Examination	01***	50%	50	Computer-Based Test	CLO01-CLO06
Total		100%					

*Continuous Evaluation (CE) is a mandatory evaluation. The CE will be based on a lab practical (file), viva, and written assessment.

**Both STs are mandatory and taken 2 times in a semester as per the Academic Schedule.

***As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

12. Syllabus of the Course:

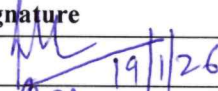
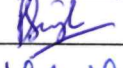
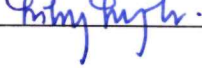
Subject: Operating System			
S.No.	Topic(s)	No. of Lectures	Weightage %
1-2	Introduction: Introduction to Operating systems, Operating System role, Computer system architecture: Single processor, Multiprocessor Systems, Clustered Systems, OS operations, Components of OS: Process management, memory management, storage management, I/O management	2	6%

3-4	OS Structure: Monolithic, Layered, Microkernel, Kernel, Shell; OS services and system calls, User and OS interface, System Calls/API, Types of System Calls, System Programs	2	6%
5-6	Introduction to Linux OS: History, philosophy, distributions, Linux architecture overview.	2	6%
7-8	Process Management Process Concept: Process definition, PCB, process states, state transition diagram. Process Operations: Process creation, termination, context switching, IPC basics.	2	6%
9-10	Threads: Overview of Threads, Multicore Programming, Multithreading Models, Threading Issues, Linux Threads	2	6%
11-13	CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithm-, First in First Out Scheduling Algorithms (FIFO), Shortest Job First Scheduling Algorithms (SJF)	3	9%
14-15	Priority Scheduling Algorithms, Scheduling Algorithms- Round-robin Scheduling Algorithms, Multilevel Queue Scheduling	2	6%
16-18	Process Synchronization: Background, The Critical-Section Problem, Two-process solution, Multiple-Process solution, Synchronization Hardware. Semaphores	3	10%
19-21	Deadlock: System Model Deadlock Characterization, Methods for handling Deadlocks, Deadlock Prevention, Deadlock avoidance, Deadlock detection, Recovery from Deadlocks	3	10%
22-25	Memory Management and Virtualisation Memory Management: Logical vs physical address space, MMU, swapping, Contiguous memory allocation, Fragmentation, Segmentation, Paging, Segmentation with paging	4	13%
26-29	Virtual Memory: Introduction, Demand Paging, Page Replacement Algorithms, Allocation of frames, Thrashing	4	13%
30-31	Mass Storage Structure: Overview, Disk Structure, Disk Attachment, Disk Scheduling, Disk Management	2	6%
32	File Management System: File Allocation methods, free-space management.	1	3%

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom, where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognise the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	Name	Signature
Course Coordinator	Dr. Chander Prabha	
Program In-Charge	Dr. Preetinder Brar	
Pro. Vice Chancellor	Dr. Jaiteg Singh	
Date (DD/MM/YYYY)	19/01/2026	